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# Demonstration of Central Limit Theorem
#Import libraries
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
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# For reproducibility
np.random.seed(42)
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# Step 1: Generate Population
population = np.random.exponential(scale=2, size=100000)
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# Step 2: Function to generate sample means
def sample_means(population, sample_size, num_samples=1000):
    means = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=sample_size, replace=True)
        means.append(np.mean(sample))
    return means
```

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# Step 3: Sample Means
means_5 = sample_means(population,5)
means_30 = sample_means(population,30)
means_50 = sample_means(population,50)
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# Step 4: Plot Results
plt.figure(figsize=(14,10))
plt.subplot(2,2,1)
sns.histplot(population,
bins=40,
kde=True)
plt.title("Original Population (Exponential)")
plt.subplot(2,2,2)
sns.histplot(means_5,
bins=30,
kde=True,
color='green')
plt.title("Sample Means (n=5)")
plt.subplot(2,2,3)
sns.histplot(means_30,
bins=30,
kde=True,
color='red')
plt.title("Sample Means (n=30)")
plt.subplot(2,2,4)
sns.histplot(means_50,bins=30,
kde=True,
color='orange')
plt.title("Sample Means (n=50)")
plt.tight_layout()
plt.show()
```

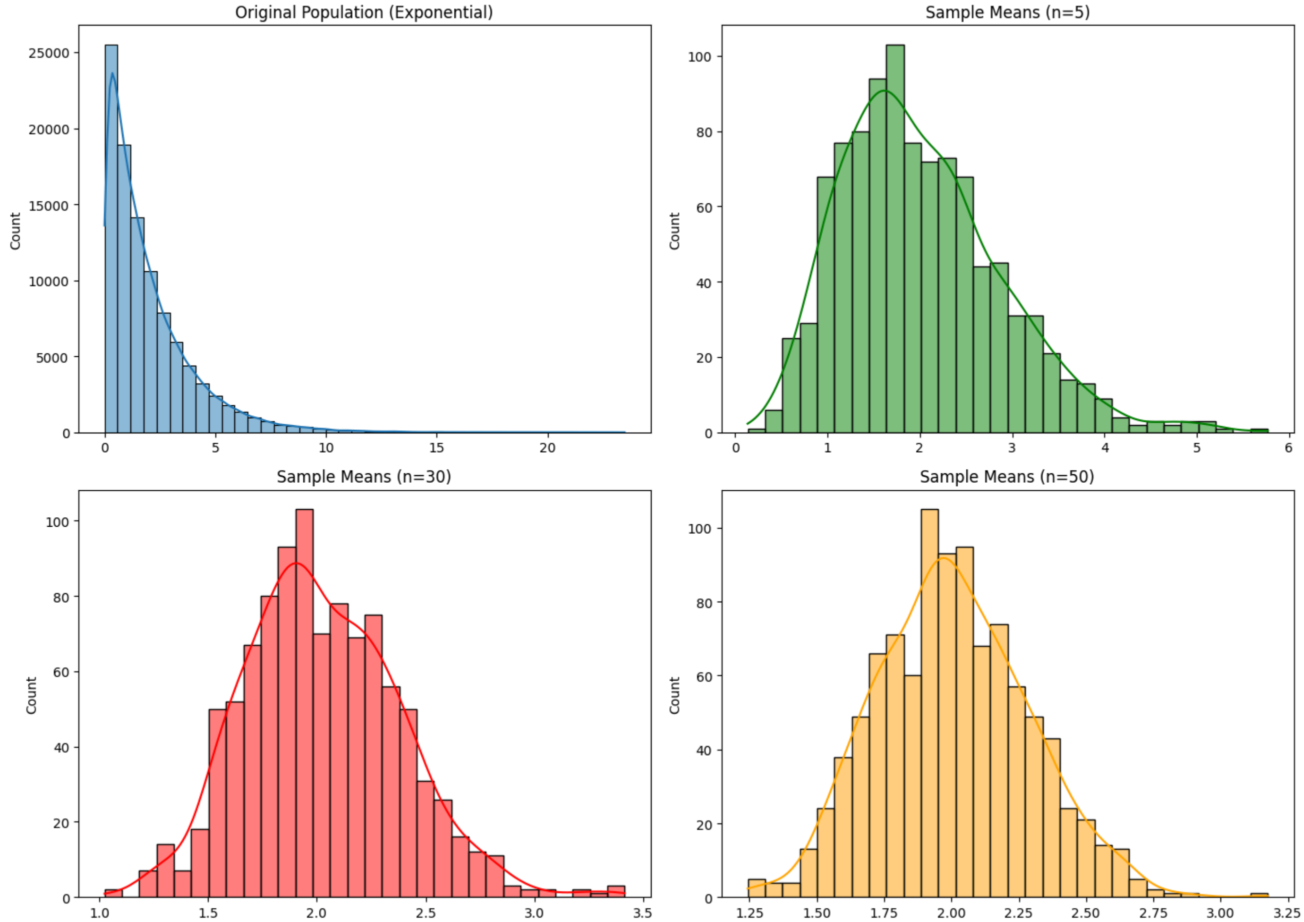
Original Population (Exponential)
Sample Means (n=5)
Sample Means (n=30)
Sample Means (n=50)
Count